

DrawVerify: Learning to Validate Circle-and-Arrow Robot Instructions from Simulator Ground Truth

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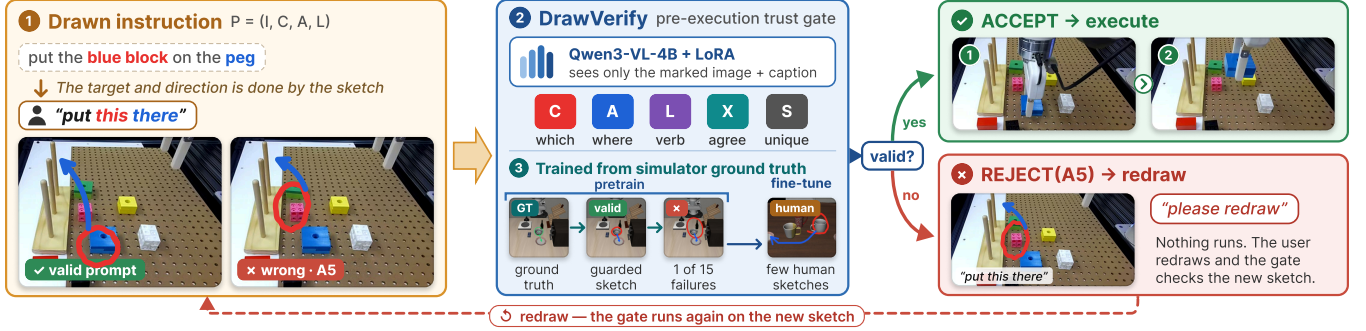


Fig. 1: Language versus sketch as an instruction interface. *Left*: even a fully formed sentence stays ambiguous when the scene contains duplicate objects: “the leftmost yellow cube” depends on the speaker’s viewpoint, and the destination must be spelled out in words. *Right*: a circle-and-arrow sketch with the name-free caption “put this there” pins down both the referent and the destination directly in image space. This paper addresses the question such an interface raises but prior work skips: how does the robot know the drawing itself is correct?

Abstract—Circle-and-arrow sketches resolve the referential and metric ambiguity that defeats language on cluttered tabletops, and unlike speech or pointing they leave a persistent artifact that can be inspected before a robot acts. Existing sketch-instruction systems consume that artifact and execute it without checking that it is correct. We formulate drawn-instruction validation as an explicit pre-execution trust gate and show that the supervision it needs is already present in a manipulation simulator. From LIBERO demonstrations we recover the manipulated object and its destination from ground-truth state, render hand-drawn-style sketches under guards that preserve validity, and apply single controlled corruptions from 15 realized failure modes, yielding 8,522 exactly labeled prompts at no annotation cost. We evaluate against 275 human sketches and a 17-participant study in which the validator gates submission live. Three results follow. Generated prompts rank validators by AUROC much as human sketches do ($\rho = +0.93$ and $+0.71$ on two human sets, comparable to the sets’ agreement with each other), although accuracy rankings do not transfer, and they inflate absolute AUROC unevenly ($+0.13$ on average against pooled human sketches). Staging generated pretraining before 82 human sketches lifts a LoRA-adapted 4B model from 0.565 to 0.808 AUROC, against 0.754 when the same rows are pooled, approaching the strongest 4-shot-prompted proprietary validator. Live gating, however, produced no detectable improvement in what participants submitted: trials the gate accepted at the second attempt were only repaired, whereas trials pushed to a third attempt broke more than they repaired, and an in-loop precision of 0.38 means most rejections interrupted a user who was already correct. Further improvement on the precision of the validator is necessary for real deployment.

Submitted to ICRA 2027 for double-blind review. Author names and affiliations are withheld in accordance with the anonymous review policy.

I. INTRODUCTION

Consider a tabletop holding three identical bowls. “Pick up the bowl” has no referent, and “put it there” presupposes an anchor that language does not carry. Language is underdetermined by construction when the distinguishing feature of a target is its position rather than its category: displacement and bearing are continuous, prepositions quantize space, and the frame in which “left” is evaluated shifts silently between and within utterances [1], [2].

Sketching addresses this directly. A user circles the object to act on, draws an arrow to the destination, and adds a name-free caption such as “move this there” [3], [4] (Fig. 1): a written form of the deictic acts people combine with speech [5], but one that does not require co-location, and so extends to teleoperation and remote supervision. The instruction is *persistent*, so it can be inspected before anything executes, and *jointly visible*, so the human’s intent and the robot’s encoding of it are the same artifact [6].

Nothing, however, inspects it. Circles miss, enclose two objects, or land on bare table; arrows invert the bearing, detach from the circled object, or contradict the caption; and channels that are individually plausible can jointly disagree. Existing sketch-instruction systems consume the drawing and act on it [3], [7], [8], tolerating noise but never classifying it. We therefore treat validation as an explicit pre-execution trust gate, in the spirit of planners that recognize their own uncertainty [9] or verify a plan before running it [10], but fired before any plan exists. Learning such a gate requires knowing the intended object and destination, which a ma-

nipulation simulator already contains. From LIBERO [11] demonstrations we recover both from ground-truth state, render hand-drawn-style sketches under guards that preserve validity, and apply single controlled corruptions from the 15 instantiated modes of an 18-mode taxonomy, yielding 8,522 exactly labeled examples at no annotation cost.

Here are our contributions:

- Drawn-instruction validation formulated as a pre-execution trust gate, with per-channel, cross-modal and sufficiency predicates, and a failure taxonomy of 18 corruption modes (15 instantiated) plus five grounding configurations (Secs. III–IV).
- A simulator-grounded pipeline giving 8,522 exactly labeled sketch prompts with single-mode failure annotations, plus 275 human sketches (Sec. V).
- Evidence that a generated benchmark orders validators like human input does, and a quantification of where it does not (Sec. VII).
- A staged training recipe that matches the AUROC of a prompted proprietary validator (Sec. VIII).
- A 17-participant in-the-loop study with a matched control gate, which detects no improvement from gating alone and points to precision, not detection, as the likely constraint (Sec. IX).

II. RELATED WORK

Sketch- and visual-prompt-based instruction. RT-Sketch [3] conditions a policy on a hand-drawn goal scene, RT-Trajectory [7] on trajectory sketches, and RoVI [4] decodes arrows and circles into 3D actions through a VLM. CrayonRobo [8], Action-Sketcher [12] and AnyUser [13] extend this to contact points, in-context reasoning and free-form user sketches. Visual-prompting methods [14] annotate images so a VLM can ground spatial decisions. All consume or produce a visual prompt and assume it is valid; CrayonRobo tolerates noisy prompts but never classifies them. We instead treat validity as the primary decision.

Deixis and verification. HRI studies how people combine speech with pointing, gaze and touch [5], resolve referring expressions through dialogue [15], and fall back on multi-modal cues when a scene contains duplicates [16]. On the verification side, KnowNo [9] asks for help when a planner is unsure, VerifyLLM [10] and VeriGraph [17] check plans against temporal logic and scene graphs, and DoReMi [18] and REFLECT [19] detect and explain execution failures. These verify *plans* or *executions*; DrawVerify verifies the *drawn instruction* before any plan is formed.

Backbones. A validator would sit upstream of a VLA policy such as OpenVLA [20] or π_0 [21], though we evaluate none here. We adapt Qwen3-VL (2B/4B) and Qwen2-VL-2B-Instruct with LoRA [22], compare against a gradient-boosting [23] geometry-only baseline (Sec. VIII-C), and build on LIBERO [11].

III. PROBLEM FORMULATION

A sketch prompt is a tuple $P = (I, C, A, L)$, where I is the RGB observation of the scene, C is the drawn circle

(center and radius) denoting *which* object to manipulate, A is the drawn arrow (tail and head) denoting *where* the object should move, and L is a short, name-free deictic caption (e.g., “move this there”). Let the scene contain ground-truth objects and regions extracted from the simulator.

We define per-channel validity predicates. $v_C(P) = 1$ iff the circle uniquely and correctly selects the intended target object (it is centered on the target, does not enclose a distractor, and is not degenerate). $v_A(P) = 1$ iff the arrow correctly specifies the destination (correct bearing and magnitude, anchored on the circled object, terminating on the intended region). $v_L(P) = 1$ iff the caption’s verb and any implicit referents are consistent with the intended action. We define a cross-modal consistency predicate $v_X(P) = 1$ iff the channels agree with one another (the circle, arrow, and caption describe the same object, destination, and predicate). Finally, a sufficiency count $N(P) \in \{0, 1\}$ indicates whether the prompt determines a *unique* executable action; $N(P) = 1$ means the prompt is sufficient (not under-determined and not ambiguous), and $v_S(P) \equiv [N(P) = 1]$.

A prompt is valid iff all conditions hold jointly:

$$V(P) = v_C(P) \wedge v_A(P) \wedge v_L(P) \wedge v_X(P) \wedge v_S(P) \quad (1)$$

The validation task is to learn a classifier $f_\theta(P) \in \{\text{ACCEPT}, \text{REJECT}\}$ that predicts $V(P)$ from the observable prompt. Crucially, a prompt can satisfy every per-channel predicate ($v_C = v_A = v_L = 1$) yet be invalid because $v_X = 0$ (channels disagree) or $v_S = 0$ (target is ambiguous), which is why validation cannot be reduced to independent per-channel checks.

IV. FAILURE TAXONOMY

Table I organizes the ways a sketch prompt can be invalid into four *corruption* families (Referential, Directional, Cross-modal, Language) plus a *Grounding* configuration family. Each corruption mode has a controlled generation mechanism and a coverage flag stating whether it is instantiated in the current release. The four corruption families define 18 modes (5 referential, 6 directional, 3 cross-modal, 4 language). Of these, 15 are generated and make up the negatives of the released dataset; the other three (C2, D1, D2) are unreachable under name-free captions. The Grounding family is counted separately from the 18: it supplies five configurations that stress *ambiguity* and *sufficiency* rather than outright error.

V. DATASET CONSTRUCTION

A. Simulator-grounded extraction

We build labels from LIBERO [11] demonstrations, whose simulator exposes object poses, region geometry, and gripper trajectories. For each demo we extract the manipulated object and its destination directly from ground-truth state, so we know, without human labeling, the correct circle target and arrow endpoints for every frame. We use 800 demonstrations spanning 40 tasks, with demo-disjoint splits so that no demonstration contributes frames to more than one split.

Family	ID	Mode	Cov.
Referential (circle / “which”)	A2	Wrong-category object	✓
	A3	Near-miss circle	✓
	A4	Empty-region circle	✓
	A5	Multi-object enclosure	✓
	A6	Degenerate size	✓
Directional (arrow / “where”)	B1	Wrong destination	✓
	B3	Wrong magnitude	✓
	B4	Wrong bearing	✓
	B5	Null destination	✓
	B6	Detached tail	✓
	B7	Spec mismatch (arrow present/absent)	✓
Cross-modal (channels disagree)	C1	Circle–arrow mismatch	✓
	C2	Caption–sketch mismatch	N/A
	C3	Predicate mismatch	✓
Language (caption / “what”)	D1	Wrong target noun	N/A
	D2	Wrong destination noun	N/A
	D3	Wrong verb	✓
	D4	Under-determined caption	✓
Grounding (ambiguity)	Five configurations: resolved; no sketch; open destination; twin referents; arrow between destinations		✓

TABLE I. The failure taxonomy. Families A–D define 18 corruption modes, of which 15 are instantiated. Cov.: ✓ generated and in the release; N/A unreachable under the name-free caption design. IDs are the generator’s internal keys and are non-contiguous.

B. Guarded human-imprecision augmentation

A perfect circle centered on the object centroid does not resemble a human drawing. We render *valid* prompts with controlled imprecision (jittered centers, non-circular strokes, imperfect radii, and hand-like arrow shafts), but we *guard* the augmentation: an in-point guard ensures the target’s keypoint remains inside a valid circle, and an out-point guard ensures distractors remain outside. This prevents “valid” augmentation from accidentally producing an ambiguous or off-target prompt, keeping labels trustworthy.

C. Controlled counterfactual failures

Negative examples are generated by applying the taxonomy of Sec. IV to valid prompts, one controlled corruption at a time, so each negative has a known failure mode and severity. This yields a balanced diagnostic signal per family rather than uncontrolled noise.

D. Name-free caption ladder

Captions are sampled from a ladder of specificity we define, L0–L3, where higher levels progressively remove object names. To force reliance on the *sketch* rather than the text, we sample only the name-free levels L2/L3 (e.g., “move this there”), consistent with the motivation that spatial grounding, not naming, should carry the instruction.

E. Composition, formats, and quality audit

Table II reports the composition. The release contains 8,522 examples (6,463 train / 900 val / 1,159 test), with 2,708 positives and 5,814 negatives. Examples are released with rendered images, raw geometric coordinates, captions, per-example labels, and failure-mode annotations. A quality audit checks guard satisfaction on positives and single-mode isolation on negatives.

Label	Train	Val	Test	Total
Valid	2,057 (31.8%)	285 (31.7%)	366 (31.6%)	2,708 (31.8%)
Invalid	4,406 (68.2%)	615 (68.3%)	793 (68.4%)	5,814 (68.2%)
Total	6,463	900	1,159	8,522

TABLE II. Dataset composition by label (train / val / test). Percentages are shares within a split; the label balance is held constant across splits.

VI. EXPERIMENTS

A. Validator training

We fine-tune three backbones with LoRA [22]: Qwen2-VL-2B-Instruct, and Qwen3-VL at 2B and 4B parameters. Adapter settings are shared: rank 32, $\alpha = 64$, dropout 0.05, applied to the query, key, value and output projections and to the gate, up and down projections of the MLP (0.93% of the parameters of Qwen3-VL-4B, 1.64% of Qwen2-VL-2B). Training is staged. The first stage learns a fresh adapter on the hand-drawn render of the generated prompts for 2 epochs, effective batch size 16, learning rate 10^{-4} with a cosine schedule and 30 warmup steps (AdamW, no weight decay, gradient clipping at 1.0). The second stage continues the same adapter on the 82 human sketches for 4 epochs, batch size 8, learning rate 3×10^{-5} . Images are fixed at 512×512 , and the loss is taken on the assistant turn only. At inference the model generates its reasoning greedily. We then force the prefix up to <answer> and read the decision from the relative probabilities of the *valid* and *wrong* tokens rather than from free-form generation. Those probabilities also supply the score for AUROC.

B. Human sketch collection

Both human datasets come from the same web interface. A participant sees a photograph of a table-top scene and a written goal, and marks the goal on the picture: a red circle around the object to act on, a blue arrow to where it should go, under a per-trial time limit. Sketches are stored as normalized stroke coordinates and rendered over the clean frame for the model.

Ungated corpus (human-275). Eleven participants produced 275 sketches over two collection rounds, covering 34 tasks across the four LIBERO suites. The interface is ungated: the participant submits once and receives no feedback, which is the baseline condition of the in-loop study below. Median drawing time was 8.2 s and 12.4% of trials were submitted by the clock. Every sketch was then labelled valid or invalid by human annotators, independently of any model; invalid sketches additionally receive a failure code from the taxonomy of Sec. IV. The corpus is naturally imbalanced at 157 valid and 118 invalid, and is split task-disjointly into 82 training and 193 held-out sketches.

In-loop study. A second cohort of 17 participants used the same interface with the validator gating submission, each scheduled for 21 trials in three blocks of seven, one per

condition, in counterbalanced order. Participants could stop early, so the realized design is incomplete: all 17 contributed no-gate and control-gate trials, but only 12 reached the model-gate block. Here the accompanying prompt is chosen from four options rather than typed, making the cross-modal channel measurable. The blocks are identical up to submission and differ only in what follows it. Under *no gate* the participant submits once and receives no feedback: the baseline interface, and the condition any gate must improve on. Under the *model gate* a fine-tuned Qwen3-VL-4B validator judges the rendered sketch and the prompt. On reject it returns “*that doesn’t look right, try again*”, up to three attempts. The *control gate* applies an identical interruption with identical wording but rejects on the ground-truth geometry of the original annotation, separating the validator’s judgement from the bare act of being sent back. The rejection carries no diagnostic in either gated condition: the failure code is logged, never shown, so the comparison is about gating rather than hints. Sketches were labelled by a human annotator from the drawing and prompt alone, shuffled into a single deck carrying no condition, attempt number or gate verdict. Of the 357 scheduled trials we exclude 107: 68 never started because participants stopped early, 11 abandoned, empty or ended by a closed browser, 16 from two goals whose written instruction was misleading and which every participant failed, and 12 drawn after the labelling deck was frozen. The analysis set is the remaining 250 trials and their 326 labelled sketches (Table VI).

VII. SYNTHETIC DATA AS A BENCHMARK

Generated prompts are cheap, and their labels are exact by construction. Human sketches are costly to collect and must be labeled by hand. Can the generated test split therefore *stand in* for human sketches when comparing validators? A benchmark does not need to reproduce a model’s absolute accuracy on human input to be useful; it needs to *order* models the way human input would. We test exactly that requirement by scoring the same seven vision-language models on generated prompts and on two independent sets of human sketches (Fig. 3), then correlating the resulting rankings.

A. Evaluation sets and protocol

Generated. 300 prompts drawn from the held-out generated test split (1,159 prompts, Table II), stratified by label only: 120 valid and 180 invalid, each sampled uniformly without replacement with a fixed seed, so the sample is 60.0% invalid against 68.4% in the full split. Families are not balanced and follow the split’s mix (invalid: 79 cross-modal or grounding, 37 language, 28 referential, 25 directional, and 11 without a released mode). No row was used for fine-tuning.

Protocol. Seven API models from three vendors (Table III) are prompted identically on all sets: the same system prompt, the same four in-context examples drawn from the human *training* split, temperature 0, and a request for both a label and a failure probability $p_{\text{fail}} \in [0, 100]$. AUROC is computed from p_{fail} with invalid as the positive class. Unparsable

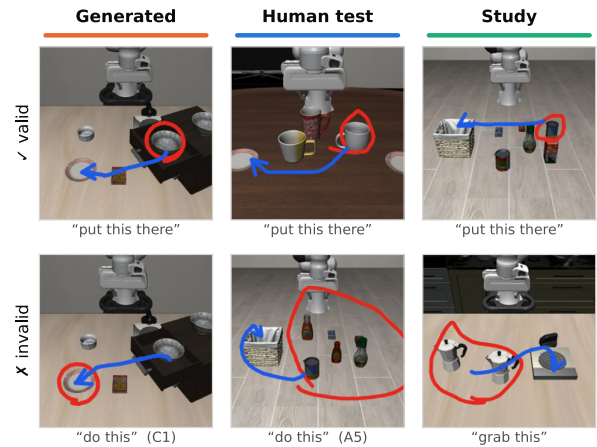


Fig. 3: The three evaluation sources, one valid (top) and one invalid (bottom) prompt each. *Generated*: the held-out test split, where the failure mode is known by construction (C1: the circle selects the plate while the arrow leaves from the bowl). *Human test*: held-out sketches of the human corpus (A5: one circle encloses several bottles). *Study*: sketches produced by participants of the human-in-the-loop study, labeled valid or invalid without a mode (the circle encloses both moka pots, so “grab this” has no unique referent).

Model	AUROC				MCC		
	Gen.	Test	Study	Pool	Gen.	Test	Study
Gemini-3.8-Flash	0.965	0.857	0.768	0.816	+0.85	+0.50	+0.43
GPT-5-mini	0.928	0.687	0.631	0.678	+0.82	+0.24	+0.25
GPT-4.1-mini	0.836	0.691	0.569	0.617	+0.64	+0.24	+0.15
Gemini-2.5-Flash	0.818	0.640	0.664	0.682	+0.65	+0.29	+0.25
Claude-Haiku-4.5	0.812	0.567	0.642	0.634	+0.61	+0.10	+0.27
GPT-5-nano	0.630	0.591	0.558	0.577	+0.31	+0.26	+0.10
Claude-3-Haiku	0.482	0.533	0.526	0.546	−0.03	+0.07	+0.10
Mean	0.782	0.652	0.623	0.650	+0.55	+0.24	+0.22
ρ vs. Gen.		+0.93	+0.71	+0.82		+0.71	+0.75
Pairs preserved		19/21	17/21	18/21		17/21	17/21

TABLE III. Seven API models on generated prompts (Gen., $n = 300$, 60.0% invalid), the human test split (Test, $n = 193$, 42.5%), the human-in-the-loop study (Study, $n = 326$, 23.6%) and both human sets pooled (Pool, $n = 519$). Bold marks the best model per column. The lower block gives each human set’s agreement with the generated ranking. (Sec. VII-A); its 6% loss on the study may shift its AUROC there.

replies are dropped (6, 4 and 20 for GPT-5-nano on the three sets; none for any other model). Ranking agreement is Spearman’s ρ over the seven models, using average ranks for ties. We also report the fraction of the 21 model pairs whose order is preserved.

B. Rankings transfer; absolute scores do not

Table III and Fig. 4 summarize the results.

The generated benchmark orders models like human sketches do. AUROC rankings on generated prompts agree with the human test split at $\rho = +0.93$, with the human-in-the-loop study at $\rho = +0.71$, and with the pooled human set at $\rho = +0.82$; 17–19 of the 21 model pairs keep their order. The agreement holds across threshold-free and class-balanced metrics: against the pooled human set, $\rho = +0.86$

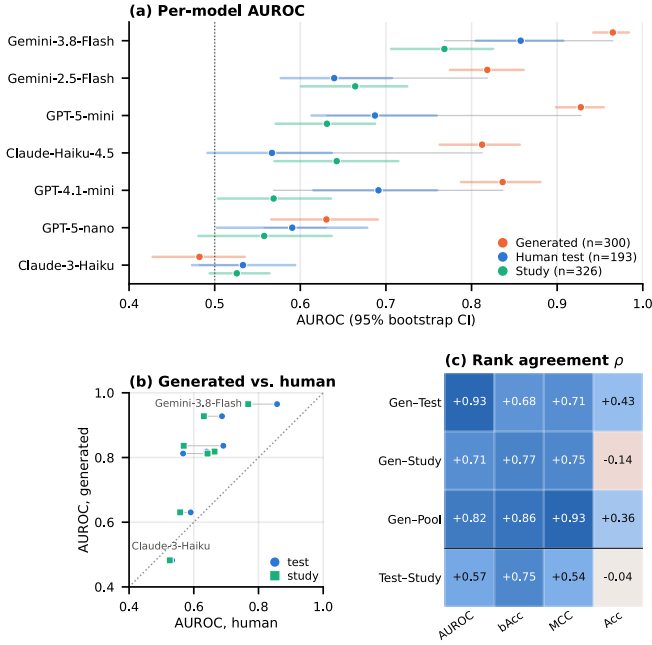


Fig. 4: Generated prompts as a benchmark. (a) AUROC of each model on generated prompts, the human test split and the human-in-the-loop study, with 95% bootstrap intervals; models are sorted by pooled human AUROC; GPT-5-nano uses the reduced counts of Table III. (b) Each model’s AUROC on generated prompts against its AUROC on the two human sets; grey segments join the two human scores of the same model. Nearly every point lies above the diagonal: generated prompts are easier, and unevenly so. (c) Spearman ρ between the seven-model rankings of two sets, per metric. The bottom row compares the two human sets with each other and serves as a reference for how much two human benchmarks agree.

for balanced accuracy and +0.93 for MCC (Fig. 4c; the underlying pooled balanced accuracy and MCC scores are not tabulated). Both ends of the ranking are stable. Gemini-3.8-Flash is first and Claude-3-Haiku last on every set. Removing any single model leaves $\rho \geq 0.71$ (AUROC) against the pooled set.

Generated agreement is comparable to human–human agreement. The two human benchmarks agree with each other at $\rho = +0.57$ (AUROC), +0.75 (balanced accuracy) and +0.54 (MCC), no higher than their agreement with generated prompts (Fig. 4c, bottom row). This does not show that the generated benchmark predicts the study ranking better than a human test set does. The difference between $\rho(\text{Gen}, \text{Study}) = +0.71$ and $\rho(\text{Test}, \text{Study}) = +0.57$ is not significant (Williams’ test for dependent correlations, $t(4) = 1.15$, $p = 0.31$), so given seven models the two are not distinguishable. We read this as evidence that generated prompts are a usable AUROC ranking instrument, not as a precise estimate of how well they rank.

Absolute scores are inflated, unevenly. Against the pooled human set, the per-model AUROC gap averages

+0.13 and ranges from -0.06 to $+0.25$ (standard deviation 0.11); it averages +0.13 against the human test split and +0.16 against the study. No constant offset converts one scale into the other, so differences of a few AUROC points on generated prompts should not be read as differences on human input.

VIII. FINE-TUNED VALIDATORS VS. PROPRIETARY MODELS

Can a small open model, adapted with LoRA, replace the strongest prompted API model, Gemini-3.8-Flash, as the trust gate, and what training data does that take? All models are scored on the identical rows from the datasets.

A. Protocol

Answer-only variants drop the rationale from the training targets. Fine-tuned scores are the forced probability of wrong over valid; decisions are the generated verdict, except for the deployed validator, which rejects at the live gate’s threshold (set for 0.85 recall on invalid sketches of the 82-sketch human training split, which yields 0.732 on the test split) and at a second cut set for 0.80 recall. Both were fixed before the study. Block (d) is the study’s no-gate arm: no validator ran, so those sketches are unassisted and scored offline. AUROC differences are paired over 2,000 bootstrap resamples.

B. Results

Staging two insufficient sources reaches Gemini-3.8-Flash. Trained on generated prompts only, Qwen3-VL-4B scores 0.565 AUROC on human sketches (block (a) of Table IV), 0.29 behind Gemini-3.8-Flash and near the zero-shot backbone (0.542), despite 0.886 on generated prompts; it transfers as indiscriminate rejection ($R_{\text{val}} = 0.495$ at 0.559 precision, against a 0.425 base rate). On the 82 human sketches alone it reaches 0.640, at 0.561 precision and 0.561 recall. Staged over those same sketches it reaches 0.808, and the gap is no longer significant (-0.049 , $[-0.107, +0.009]$; seeds 0.810 ± 0.003). With the human stage cut to 40 sketches it still reaches 0.847, beating 82 unstaged sketches by 0.21 AUROC. Order matters as much as composition: one shuffled pool over the same rows gives 0.754 (-0.10^*). Both 2B backbones stay behind even when staged.

Parity holds on sketches no validator saw. On the full study the staged validator scores 0.730 against 0.768 (-0.038 , $[-0.095, +0.020]$), but nearly three quarters of that set comes from gated arms where participants could adapt to the gate. On the 88 ungated sketches it scores 0.830 against 0.797, at higher accuracy (0.818 vs. 0.761) and higher R_{val} (0.824 vs. 0.743); with 14 invalid sketches this supports parity, not an advantage. The two models sit at very different operating points, Gemini-3.8-Flash catching 96.3% of invalid test sketches while rejecting half the valid ones ($R_{\text{val}} = 0.505$), so AUROC parity does not imply parity as a deployed gate. The 0.80-recall cut leaves AUROC untouched and gives the best accuracy and MCC in block (d). The comparison is also asymmetric: the validator is fine-tuned,

Model	Acc	Prec	Rec	MCC	R _{val}	AUROC	$\Delta_{3.8}$
<i>(a) Human test split (n = 193, 42.5% invalid)</i>							
Gemini-3.8-Flash	0.699	0.590	0.963	+0.50	0.505	0.857	
Qwen3-VL-4B zero-shot	0.585	0.600	0.073	+0.08	0.964	0.542	−0.32*
Qwen3-VL-4B gen	0.653	0.559	0.866	+0.38	0.495	0.565	−0.29*
Qwen3-VL-4B human only	0.627	0.561	0.561	+0.24	0.676	0.640	−0.22*
Qwen3-VL-4B mixed	0.684	0.598	0.780	+0.39	0.613	0.754	−0.10*
Qwen3-VL-4B gen→hum	0.798	0.779	0.732	+0.58	0.847	0.808	−0.05
answer-only	0.808	0.817	0.707	+0.61	0.883	0.841	−0.02
40 human sketches	0.782	0.900	0.549	+0.57	0.955	0.847	−0.01
Qwen3-VL-2B gen→hum	0.762	0.757	0.646	+0.51	0.847	0.779	−0.08*
Qwen2-VL-2B gen→hum	0.741	0.767	0.561	+0.46	0.874	0.720	−0.14*
<i>(b) Generated prompts (n = 300, 60.0% invalid)</i>							
Gemini-3.8-Flash	0.927	0.899	0.989	+0.85	0.833	0.965	
Qwen3-VL-4B gen	0.840	0.800	0.978	+0.68	0.633	0.886	−0.08*
answer-only	0.930	0.908	0.983	+0.86	0.850	0.943	−0.02
Qwen3-VL-4B mixed	0.910	0.901	0.956	+0.81	0.842	0.897	−0.07*
Qwen3-VL-2B gen	0.823	0.777	0.989	+0.65	0.575	0.880	−0.09*
Qwen2-VL-2B gen	0.863	0.880	0.894	+0.71	0.817	0.787	−0.18*
<i>(c) Human-in-the-loop study (n = 326, 23.6% invalid)</i>							
Gemini-3.8-Flash	0.739	0.468	0.766	+0.43	0.731	0.768	
Qwen3-VL-4B gen→hum	0.745	0.466	0.532	+0.33	0.811	0.730	−0.04
0.80-recall threshold	0.742	0.456	0.468	+0.29	0.827	0.730	−0.04
answer-only	0.730	0.432	0.455	+0.27	0.815	0.720	−0.05
<i>(d) Ungated study sketches, scored offline (n = 88, 15.9% invalid)</i>							
Gemini-3.8-Flash	0.761	0.387	0.857	+0.46	0.743	0.797	
Qwen3-VL-4B gen→hum	0.818	0.458	0.786	+0.50	0.824	0.830	+0.03
0.80-recall threshold	0.830	0.478	0.786	+0.52	0.838	0.830	+0.03
answer-only	0.773	0.364	0.571	+0.32	0.811	0.767	−0.03
<i>(e) Recall on invalid prompts by failure family</i>							
	Ref.	Dir.	X-mod./Gr.	Lang.	Withh.	All	R _{val}
Human test split (82 invalid: 50 / 19 / 13)							
Gemini-3.8-Flash	0.96	0.95	1.00			0.963	0.505
Qwen3-VL-4B gen→hum	0.80	0.53	0.77			0.732	0.847
Generated prompts (180 invalid: 28 / 25 / 79 / 37 / 11)							
Gemini-3.8-Flash	1.00	1.00	0.99	0.97	1.00	0.989	0.833
Qwen3-VL-4B gen	1.00	0.96	0.97	1.00	0.91	0.978	0.633

TABLE IV. Fine-tuned validators against Gemini-3.8-Flash on identical rows. Positive class is INVALID; Prec and Rec are computed on invalid prompts, R_{val} is recall on valid ones. $\Delta_{3.8}$ is the paired AUROC difference to Gemini-3.8-Flash, * marking a 95% bootstrap interval excluding zero. Block (e) splits recall on invalid prompts by failure family (X-mod./Gr.: cross-modal modes C1 and C3 together with invalid Grounding configurations; Withh.: invalid prompts that carry no released failure-mode label), with invalid counts per column in its sub-headers; only the human test split and generated prompts carry failure modes. Recipes and thresholds are defined in Sec. VIII-A; the 0.80-recall cut was evaluated only on study sketches, blocks (c) and (d).

while Gemini-3.8-Flash receives four in-context examples. We ran no many-shot or fine-tuned API condition, so this is parity with what prompting achieves, not with what the proprietary model could reach.

The human-sketch gap is directional. On generated prompts both models are near ceiling in every family (block (e) of Table IV), but on human sketches the staged validator trails Gemini-3.8-Flash most on directional failures (0.53 vs. 0.95), against 0.80 vs. 0.96 on referential and 0.77 vs. 1.00 on the 13 cross-modal sketches. Gemini-3.8-Flash buys that recall by accepting only half of the valid sketches.

C. Geometry-only baselines

Generated failures perturb stroke coordinates, so geometry alone may suffice. We segment the circle and arrow out of each rendered image by colour, compute features from

Model	Test (a)	Study (c)	Ungated (d)
Geometry, gen	0.741	0.531	0.503
Geometry, hum	0.640	0.615	0.421
Geometry, gen+hum	0.736	0.557	0.540
Gemini-3.8-Flash	0.857	0.768	0.797
Qwen3-VL-4B gen→hum	0.808	0.730	0.830

TABLE V. Pixel-geometry baselines. Bold marks the best geometry fit per column. Every geometry AUROC is significantly below Gemini-3.8-Flash (95% paired bootstrap).

Condition	Tr.	Sk.	First	Final	Att.	Fix/Brk	Cap 2
No gate	88	88	0.841	0.841	1.00	0/0	0.841
				[.750, .903]			[.750, .903]
Model gate	78	108	0.782	0.795	1.38	5/4	0.821
				[.692, .870]			[.721, .890]
Control gate	84	130	0.714	0.726	1.55	3/2	0.762
				[.623, .810]			[.661, .840]
Total	250	326					

TABLE VI. Validity of the instruction finally sent, by condition, and the same loop replayed with a one-retry cap. Fix/Brk counts trials whose validity changed between the first attempt and what was sent. Tr.: trials; Sk.: sketches labelled; Att.: mean attempts per trial; Cap 2: the same loop replayed with at most one retry. Wilson 95% intervals.

pixel positions only (centroid, radius, arrow endpoints, tail and tip relative to the circle), and fit a gradient-boosting classifier [23] without the caption (Table V). Fits that use generated data reach 0.74 AUROC on the human test split but only 0.53–0.56 on the study and 0.50–0.54 on ungated sketches, and every fit is significantly below Gemini-3.8-Flash on every set: stroke geometry alone does not validate human sketches.

IX. RESULTS OF THE HUMAN EXPERIMENT

The study asks whether interposing a validator between the participant and the model improves the sketch that finally gets sent. Section IX-A reports that we detect no improvement, and shows even opposing effects of shallow and deep retries. Section IX-B points to precision rather than detection as the likely cause, and traces the residual failures to the interface.

A. No detectable effect of the loop

a) The study detects no improvement: No gated condition beats the ungated baseline (Table VI). Paired within participant, the model gate scores −0.075 against no gate (95% CI [−0.196, +0.046]), better for 4 of the 12 participants who reached the model-gate block and worse for 6, at a cost of 1.38 attempts per trial against 1.00. The contrast that isolates the validator from the bare act of interruption, model gate against control gate on the same 12 participants, is +0.060 (95% CI [−0.038, +0.157]; model better for 5, worse for 4, tied for 3; sign test $p = 1.00$). No contrast is significant, but the design cannot rule out modest effects: the paired model-gate vs. no-gate interval admits gains of up to 4.6 points, and from its standard error the minimum difference detectable at 80% power ($\alpha = 0.05$) is about 17

Condition	Att.	Tr.	Sk.	First	Final	Fix/Brk
Model gate	1	56	56	0.839	0.839	0/0
	2	14	28	0.500	0.786	4/0
	3	8	24	0.875	0.500	1/4
	all	78	108	0.782	0.795	5/4
Control gate	1	53	53	0.774	0.774	0/0
	2	16	32	0.500	0.625	2/0
	3	15	45	0.733	0.667	1/2
	all	84	130	0.714	0.726	3/2

TABLE VII. Outcomes by the attempt at which the trial ended. Att.: the attempt the trial ended on; Tr., Sk., Fix/Brk as in Table VI. Trials accepted at the second attempt were only repaired; trials that reached a third broke more than they repaired.

Sketches scored	Sk.	Inv.	Prec	Rec	Spec
<i>Validator</i>					
Ungated arm (unselected)	88	14	0.458	0.786	0.824
Model arm (loop-selected)	108	26	0.382	0.500	0.744
Control arm (loop-selected)	130	37	0.567	0.459	0.860
All	326	77	0.466	0.532	0.811
<i>Control gate (geometry rule)</i>					
Its own arm	130	37	0.362	0.568	0.602

TABLE VIII. The deployed Qwen3-VL-4B validator, at the live threshold, against the human labels by arm. Positive class is INVALID. Gated arms are loop-selected: every sketch after a first attempt exists because a gate rejected the one before it. The three arm rows partition the 326 sketches, and the All row is block (c) of Table IV. The last row scores a different decider, the control gate’s geometry rule, on its own arm.

points. We therefore read the result as an absence of evidence for a benefit, not as evidence that gating has none.

b) Shallow and deep retries pull in opposite directions:

Table VII separates two effects that the arm-level averages hide. Among trials the gate accepted at the second attempt, validity only improved (6 repaired, none broken). That row is conditioned on acceptance: the same second attempt broke at least one model-gate trial that the gate then rejected again. Trials that ran to a third attempt broke more than they repaired: they began mostly valid and ended at 0.500 and 0.667, with 6 broken against 2 repaired. In these few trials a first rejection usually found a genuine mistake, while a third more often contradicted a participant who had been right initially.

c) A one-retry cap is a hypothesis, not a finding:

If the damage is concentrated in the deep tail, truncating it should recover the loss, and replaying the logged attempts is consistent with this. Under a one-retry cap the last column of Table VI gives 0.821 [.721, .890] and 0.762 [.661, .840], against 0.795 and 0.726 for the three-attempt loop as run and 0.782 and 0.714 for no retry at all. The intervals overlap almost entirely, and the difference between the cap and the full loop rests on the 8 model-gate and 15 control-gate trials that reached a third attempt. The replay also holds behaviour fixed while changing the budget: a participant who knew that only one retry was available would plausibly revise more carefully. We therefore offer the cap as a hypothesis for a follow-up study.

B. Why: precision, not detection

a) Precision, not detection, is the likelier bottleneck:

The gate’s in-loop numbers look poor (precision 0.382, recall 0.500; Table VIII) and may partly reflect selection: on the model arm every sketch after a first attempt exists *because* the validator rejected the one before it, so it partly chooses its own test cases. Selection alone does not explain them, however. The control arm is also loop-selected, by the geometry rule, yet there the validator’s precision is its highest (0.567) and its recall its lowest (0.459). Scored offline on the 88 ungated sketches, which it had no hand in producing, its recall rises to 0.786 at specificity 0.824 and AUROC 0.830, above the 0.808 measured on held-out corpus sketches. Ranking ability transfers to human drawings; precision does not. On the model arm, where its rejections actually interrupted someone, precision is 0.382, so more than 60% of in-loop rejections interrupted a participant who had already succeeded. On sketches it did not gate it reaches only 0.458 (ungated arm) and 0.567 (control arm), and 0.466 over all 326. That cost is what the second-attempt repairs spend themselves against, and it is a threshold problem rather than a detection failure.

b) The residual failures are an interface effect: Of the six trials broken at a third attempt, five changed the *prompt* rather than the drawing. Told only that something was wrong, participants revise the cheapest element on screen, replacing a correct specific phrase with the generic “*put this there*”, which then contradicts what they drew. This is a property of the feedback we gave, not of the model: naming the rejected channel (drawing or words) without naming the fix should recover most of the loss.

X. LIMITATIONS

Coverage and domain. Three of the eighteen corruption modes (C2, D1, D2) are not instantiated, because name-free captions carry no noun to corrupt. Physical failures such as occluded targets are outside the scope of this work. All data comes from LIBERO tabletop scenes, with human sketches drawn over rendered frames of those same scenes. We use no real camera imagery, and we never execute an accepted prompt, so validity is scored against human judgments of the drawing rather than task success.

Statistical power. The ranking-transfer result rests on seven models, too few for a precise estimate of ρ , and block (d) of Table IV contains only 14 invalid sketches: enough for parity with Gemini-3.8-Flash, not for an advantage. With 12 participants in the model-gate block the study’s minimum detectable effect is about 17 points, and the one-retry cap rests on 23 three-attempt trials.

Human-likeness is not measured. We never compare generated stroke statistics with the human corpus or test whether people can tell them apart. The uneven AUROC inflation suggests the two populations differ.

Baseline and gate design. Gemini-3.8-Flash receives four in-context examples while our validators are fine-tuned; we ran no many-shot or fine-tuned API condition, so the reported parity is with prompting alone. The deployed gate is binary

and its rejection carries no diagnostic, which isolates gating from hinting but produces the dominant failure of Sec. IX. A per-channel verdict was never tested.

XI. CONCLUSION

We treated the drawn instruction itself as the object of verification, rather than the plan it produces, and showed that a manipulation simulator supplies the supervision this requires without human labeling. The resulting 8,522-prompt corpus, its failure taxonomy (15 of 18 modes instantiated), and 275 human sketches are released with the code.

Generated prompts inflate absolute scores unevenly against pooled human sketches, by +0.13 AUROC on average and from -0.06 to $+0.25$ per model. Accuracy rankings do not transfer either, so generated prompts cannot substitute for human evaluation, and operating points must be set on human input. By AUROC and class-balanced metrics, however, they order validators much as human sketches do. Staging matters more than volume: the same rows give 0.754 AUROC pooled and 0.808 staged. Moreover, 40 staged sketches beat 82 unstaged by 0.21. A LoRA-adapted open 4B validator matching a 4-shot-prompted proprietary model on sketches it never influenced (0.830 vs. 0.797) suggests an on-device trust gate is practical.

Yet putting that gate in front of participants produced no detectable improvement. That null hides two opposing effects: trials accepted after one retry gained six valid sketches and lost none, while third attempts broke six and repaired two. Precision, not detection, looks like the binding constraint: inside the loop only 0.38 of rejections were correct.

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